

## Day 2 Of Communication System Class

### Performance Parameters/Impairments of Communication System

#### 1. Noise

- Definition: Unwanted electrical signal that interferes with the original message.
- Measurement: Signal-to-Noise Ratio (SNR).
- Note: Noise is always present; the goal is to maximize SNR.

#### 2. Distortion

- Definition: Unwanted change in the shape or characteristics of the transmitted signal.
- Effect: Received signal differs from the transmitted signal.
- Mark in notes: Often marked with  $X$  or  $(\varphi)$ .

#### 3. Delay (Latency)

- Components:
- Processing Delay: Time taken by devices to process the signal/packet.
- Propagation Delay: Time for the signal to travel through the medium (depends on distance and medium speed).
- Jitter: Variation in packet delay (inconsistent delays).
- Overall: Called Latency.

#### 4. Data Rate / Baud Rate / Throughput

- Data Rate: Number of bits transmitted per second (bps).
- Baud Rate: Number of signal changes (symbols) per second.
- Throughput: Actual successful/ effective data transmission rate (usually  $\leq$  data rate due to overhead, errors, etc.).

## 5. Bandwidth (BW)

- **Analog Signal:**

$$BW = f_{\max} - f_{\min} \quad (\text{in Hz})$$

- **Digital Signal:** Depends on sampling.

**Nyquist Sampling Theorem:**

$$f_s = 2 \times f_{\max}$$

- **PCM Bit Rate:**

$$\text{Bit Rate} = n \times f_s = n \times 2 \times f_{\max}$$

where  $n$  = number of bits per sample.

**Examples:**

- **FM Radio:** 88 MHz – 108 MHz (BW = 20 MHz).
- **Voice Signal:** 300 Hz – 3400 Hz  $\approx$  **4 kHz** (with guard band consideration).
- **Voice PCM:**  
 $8 \times 2 \times 4000 = 64,000 \text{ bps} = 64 \text{ kbps}$  (standard telephone quality).

## 6. Channel Capacity (CC) – Shannon-Hartley Theorem

According to **Claude Shannon**:

$$C = B \log_2 \left( 1 + \frac{S}{N} \right)$$

Where:

- $C$  = Channel Capacity (bps)
- $B$  = Bandwidth (Hz)
- $\frac{S}{N}$  = Signal-to-Noise Ratio (linear, not dB)
- Assumes **Gaussian noise**.

**Note:**  $S$  = Signal power,  $N$  = Noise power.

## 7. Signal-to-Noise Ratio (SNR)

- In dB:  
$$\text{SNR (dB)} = 10 \log_{10} \left( \frac{S}{N} \right)$$
- Interpretation:
  - Positive (+): Signal > Noise (good).
  - Negative (-): Signal < Noise (poor).
- Linear form:  $\frac{S}{N}$ .

### [ Most Probable Question ]

1. Calculate the theoretical channel capacity of a communication link if its bandwidth is 4000 Hz and its operating SNR is 20 dB.

**Solution:**

1. Convert SNR to linear:

$$\frac{S}{N} = 10^{20/10} = 10^2 = 100$$

2. Apply Shannon formula:

$$C = 4000 \times \log_2(1 + 100) = 4000 \times \log_2(101)$$

3.  $\log_2(101) \approx 6.6582$

$$C \approx 4000 \times 6.6582 = 26,632.8 \text{ bps} \approx 26.63 \text{ kbps}$$

**Final Answer:**

$$C \approx 26.63 \text{ kbps}$$